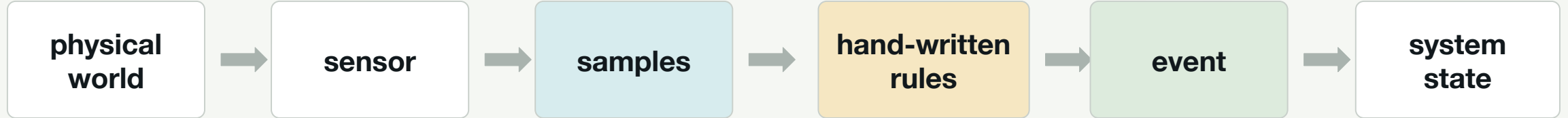


Learning patterns from examples

When writing reliable rules is the hard part

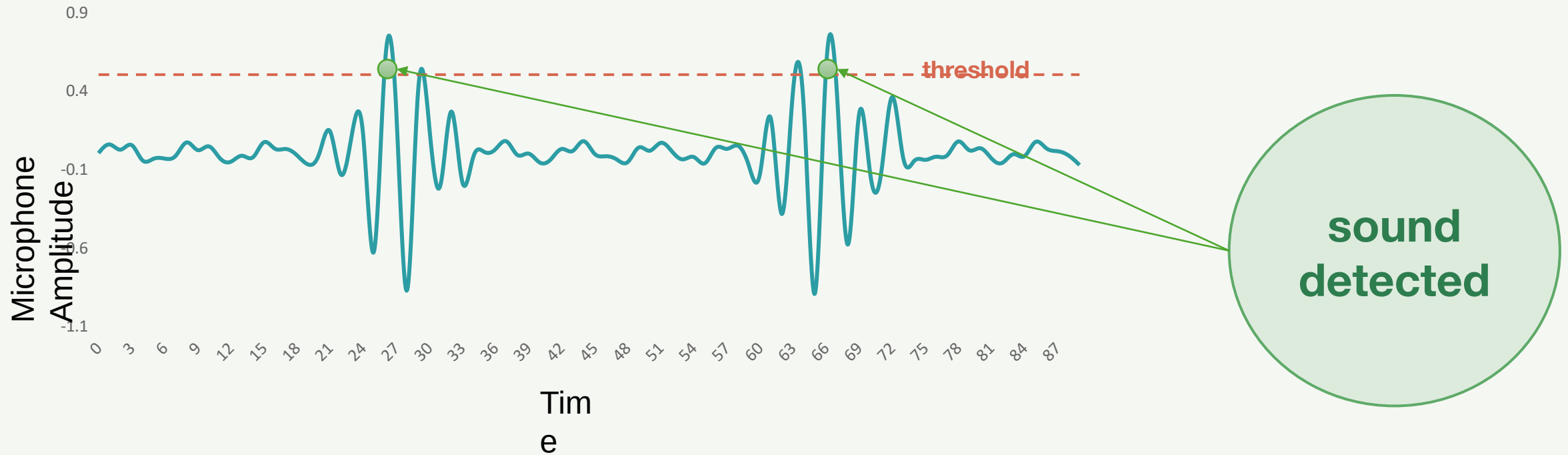


Sensor-to-Action Pipeline



A threshold can tell us that sound is "present"

but nothing about "what it is"



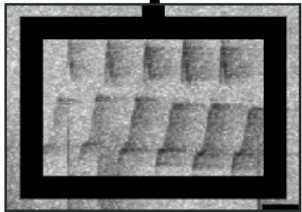
But what is it?

What kind of bird made the sound?

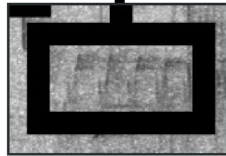
What could we measure from the recording?



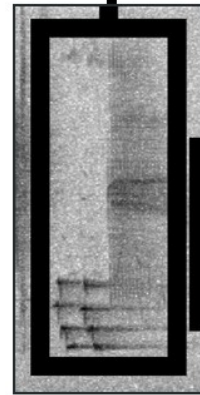
Spectrograms



Black-and-white
Warbler



Kentucky
Warbler



Wood
Thrush

TIME

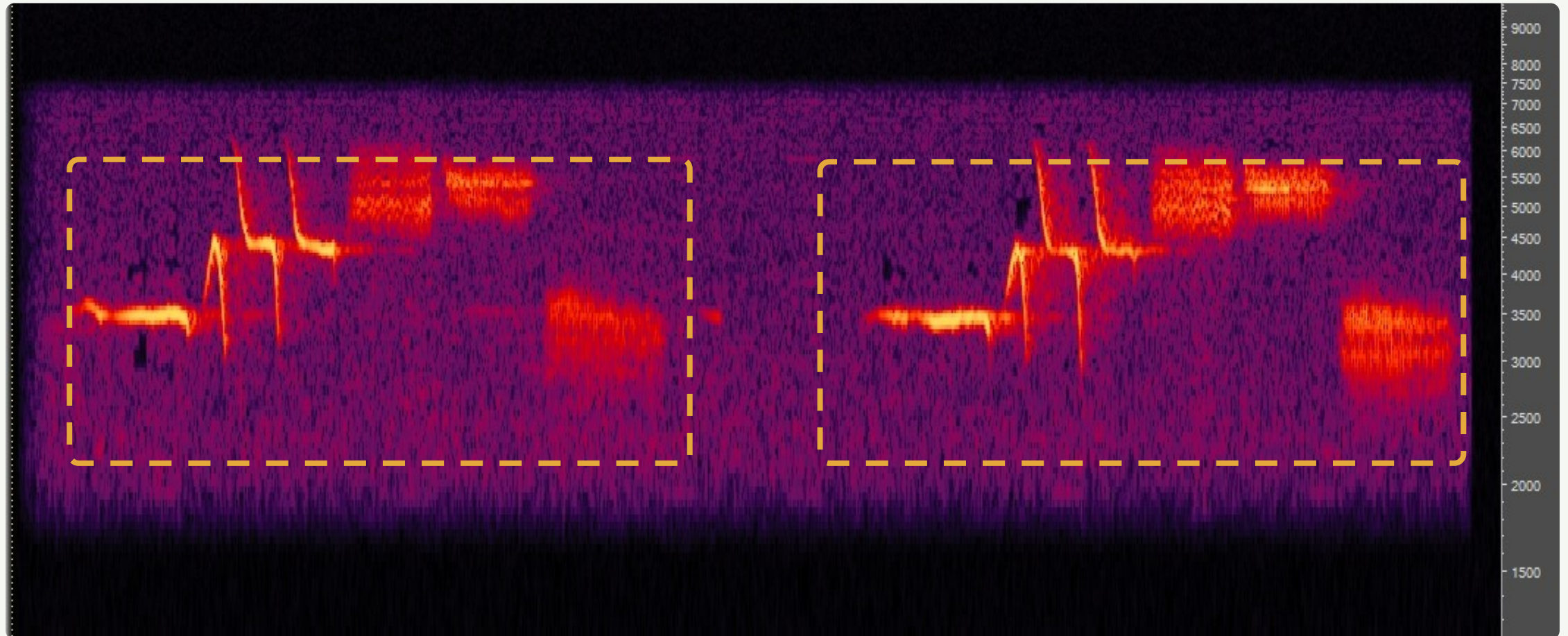
left → right

FREQUENCY

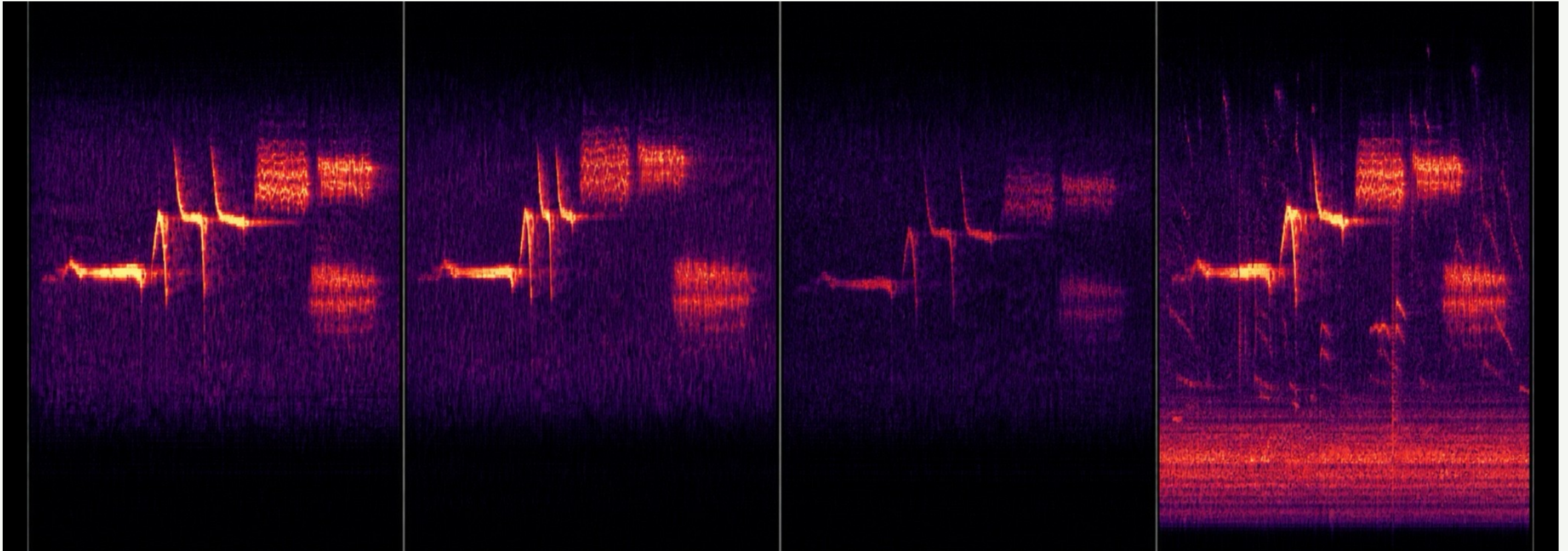
low → high

**Darker marks = stronger
energy**

Rule-Based Pattern Matching



Within-Class Variation



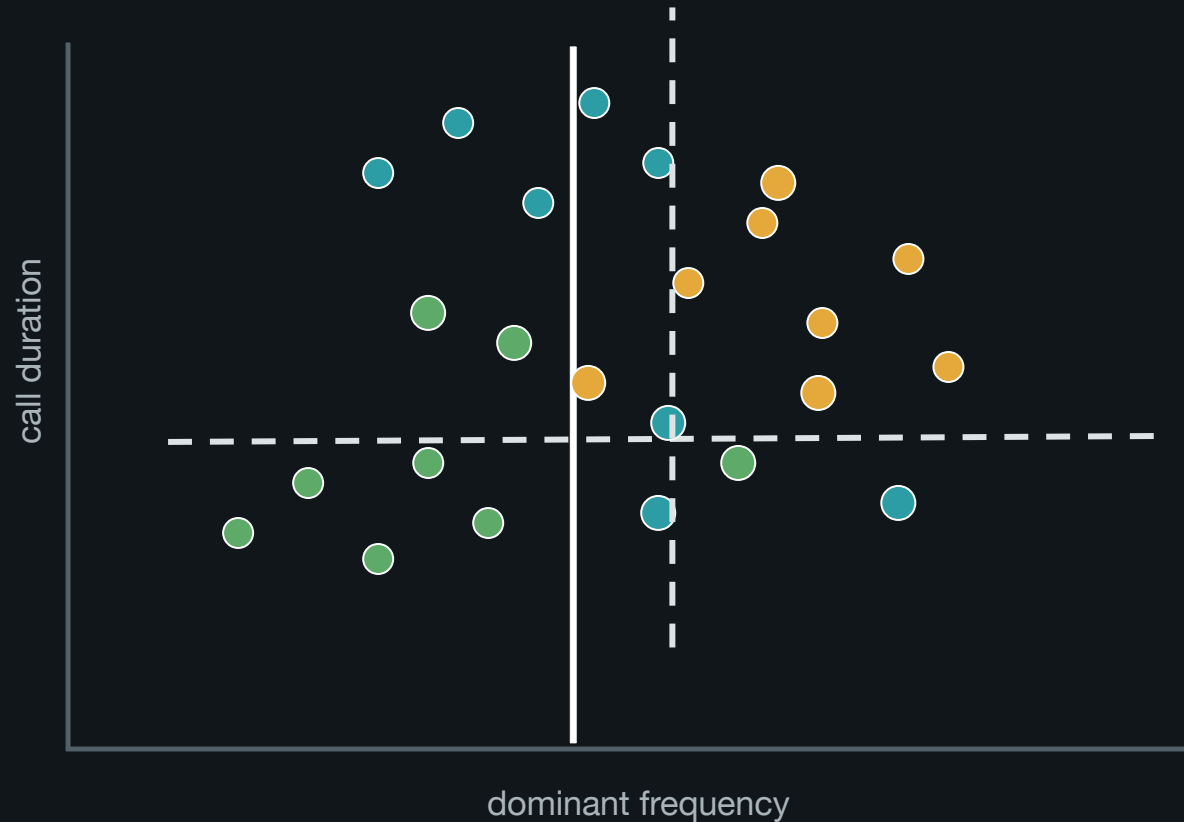
another individual

different timing

farther from the
microphone

wind + other sounds

Hand-Crafted Decision Rules

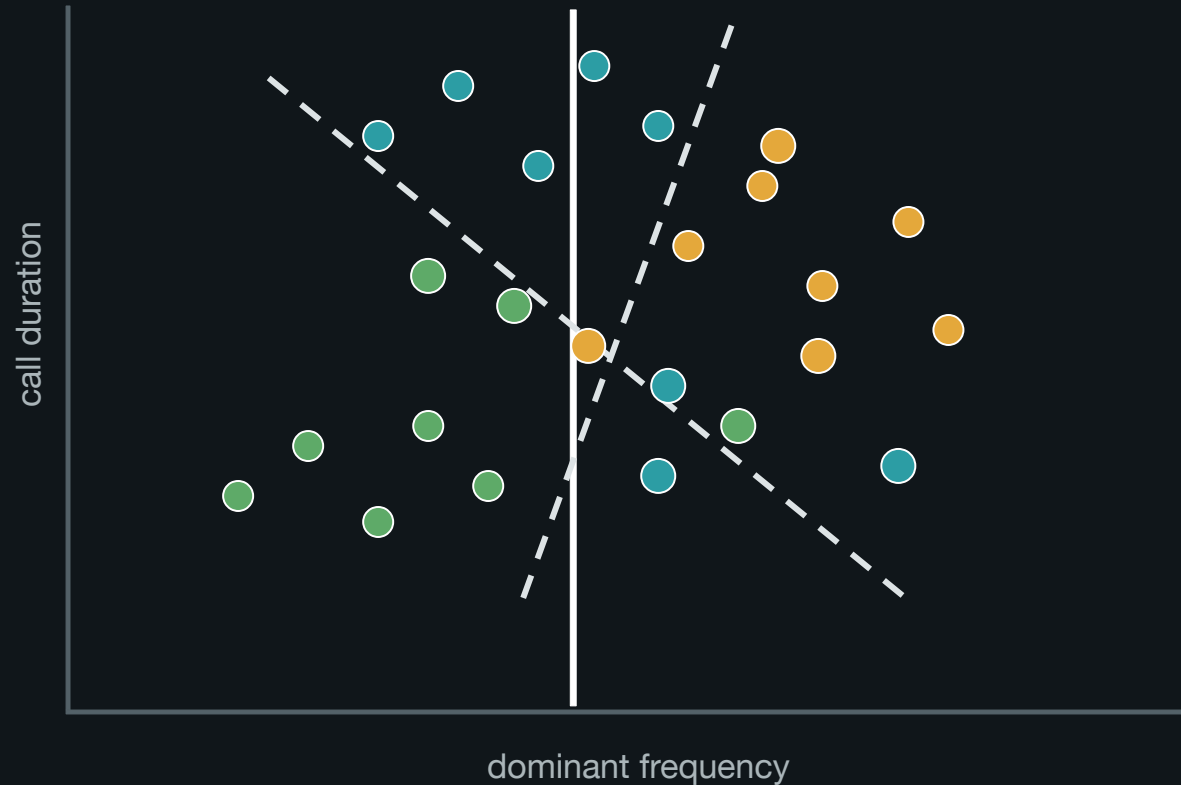


if pitch > ...

And the call is shorter than ...

except when wind adds ...

Feature Engineering



if pitch * call_duration > ...

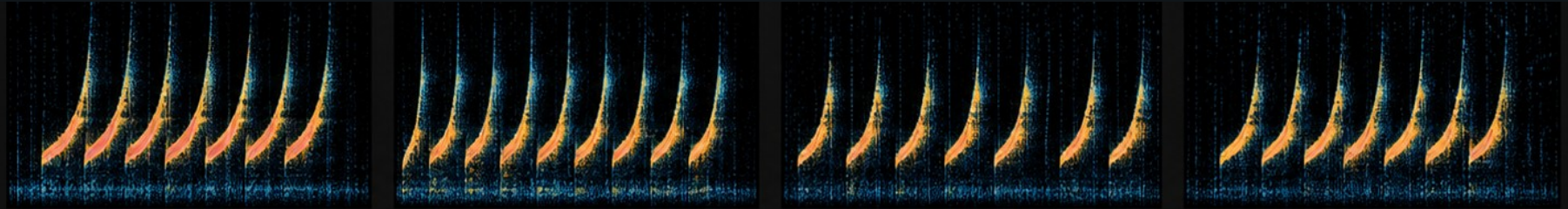
unless the call is faint with xyz ...

except when wind adds ...

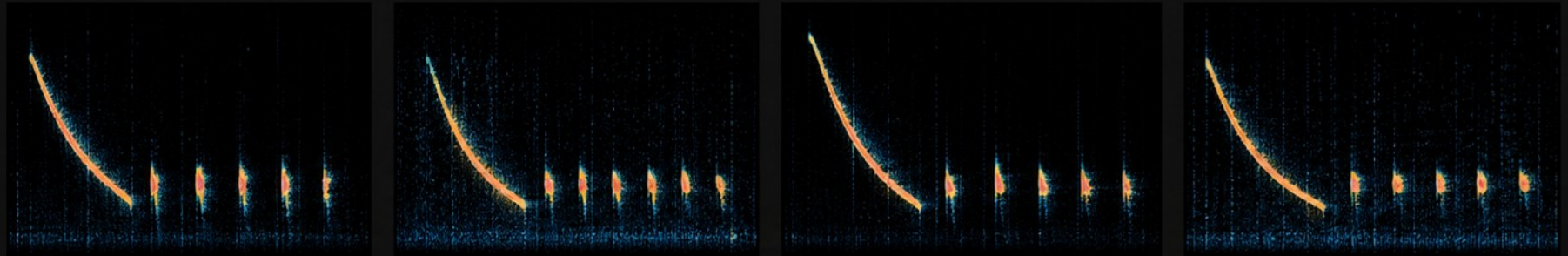
**Hand-crafted rules only work
when the patterns are very
consistent**

Supervised Learning from Labeled Examples

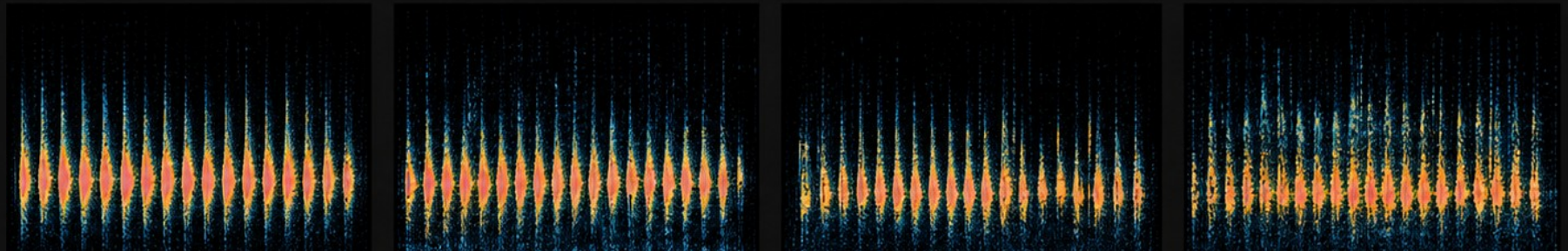
SPECIES A



SPECIES B

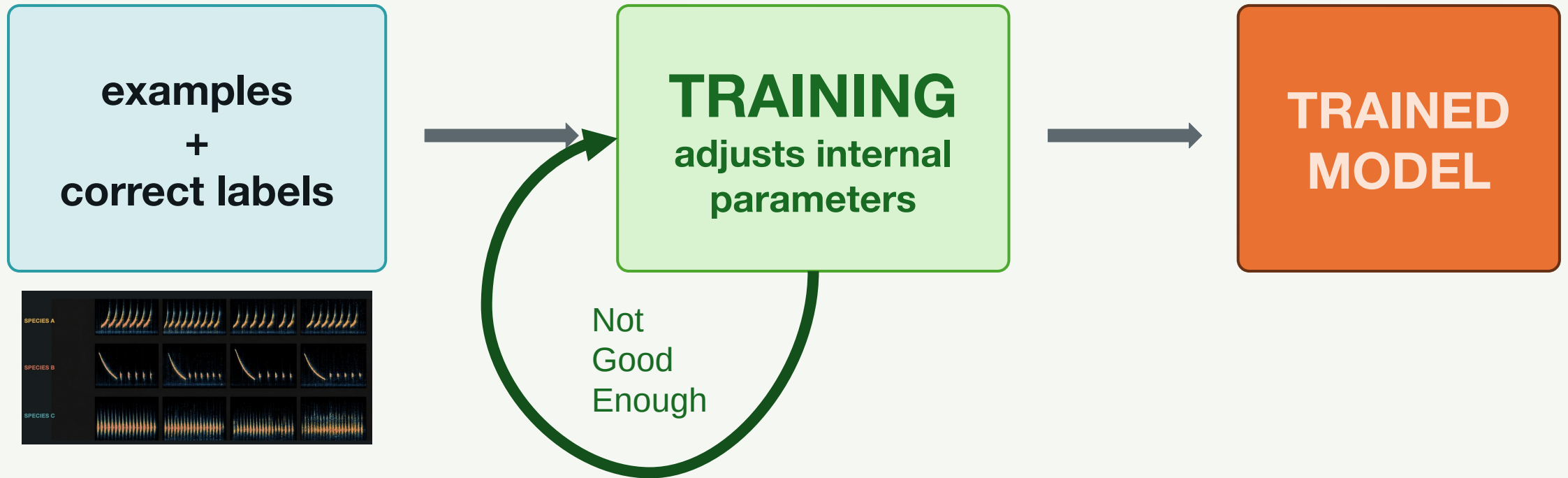


SPECIES C



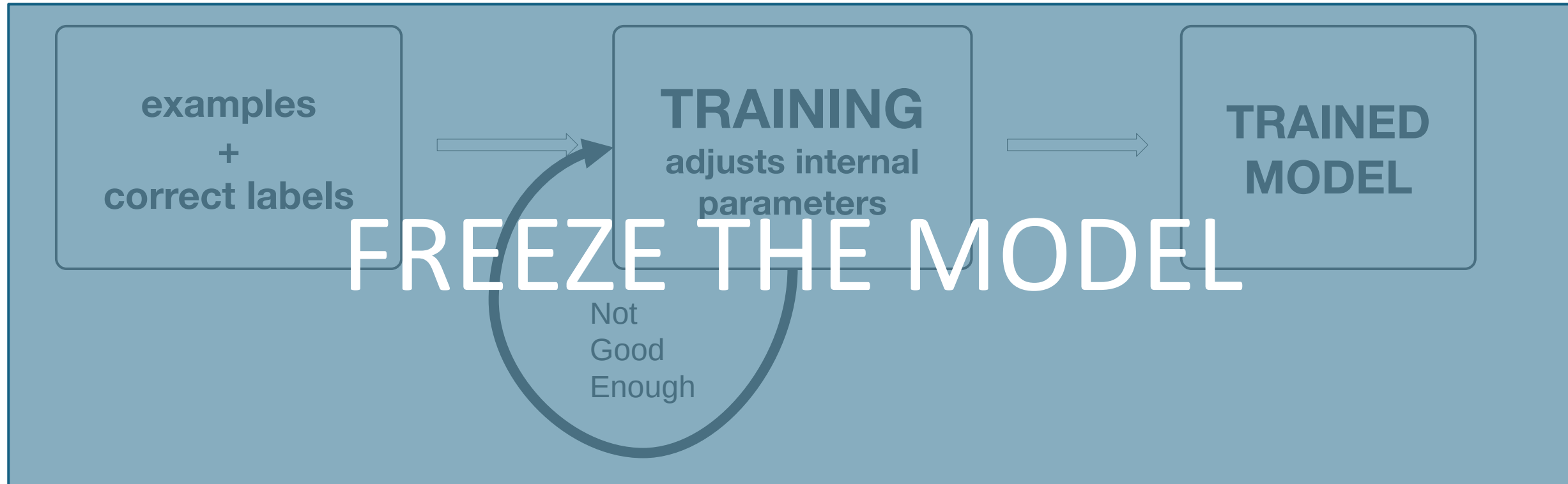
same label $\neq$ identical signal

Model Training



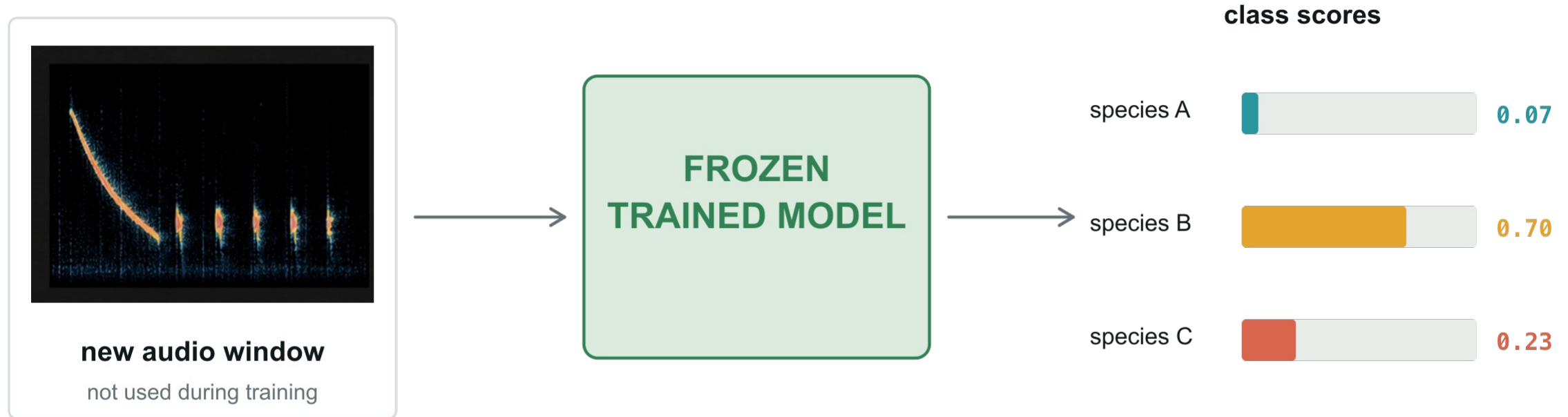
class = category | label = assigned class

Frozen Model



We freeze the model once we're "happy" with it

Inference



The model is unchanged. Only the input is new.

Training and Inference

TRAINING

INPUT: many examples + correct labels

LEARNING: adjust parameters

OUTPUT: produce one fixed model

learn once during development

INFERENCE

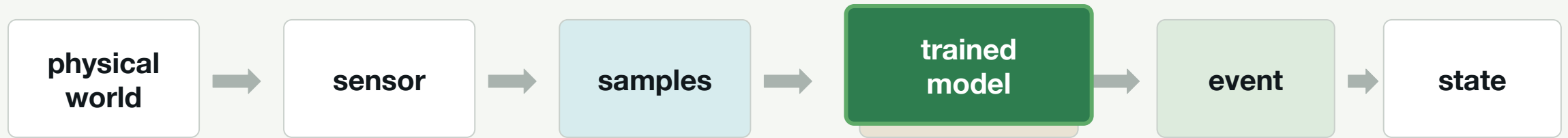
INPUT: a new sample + fixed trained model

INFERENCE: compute score

OUTPUT: produce one prediction
(aka: the label for this sample)

run whenever new data arrives

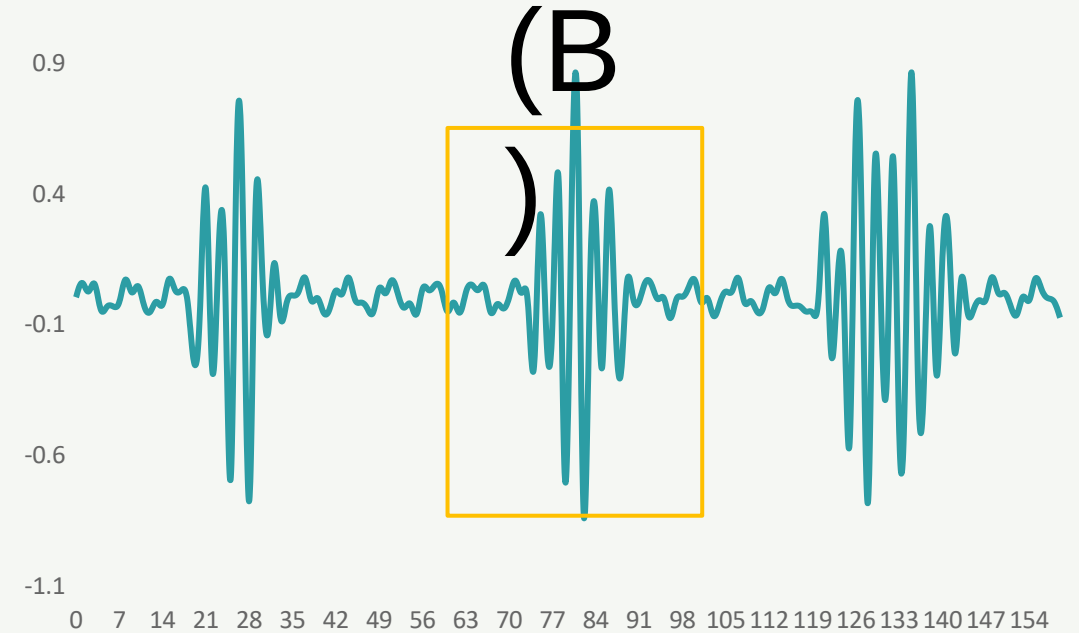
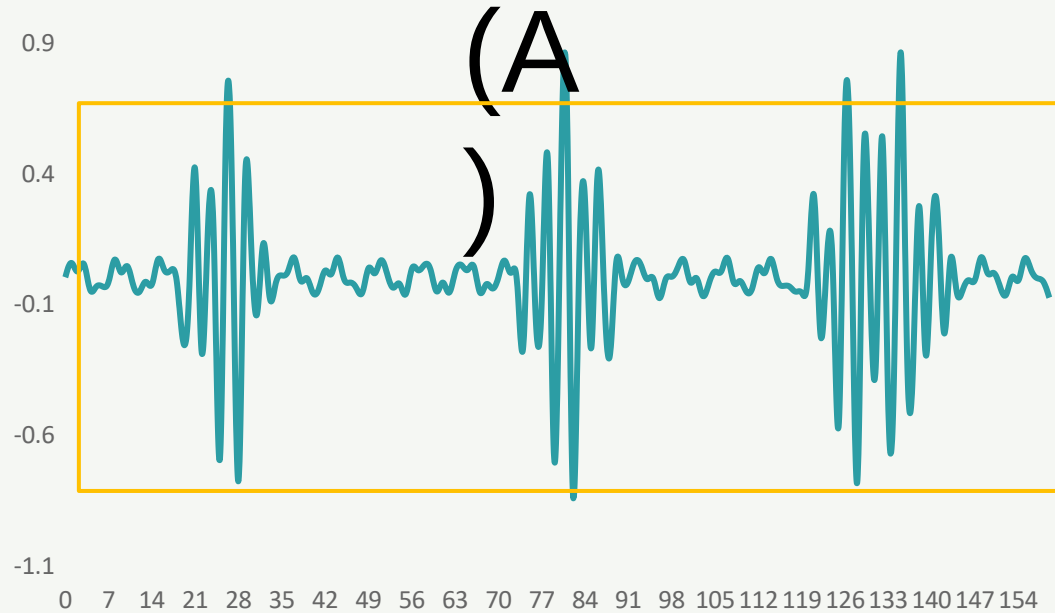
Learned Model in a Sensor Pipeline



Same sensing pipeline. Different way to interpret the samples.

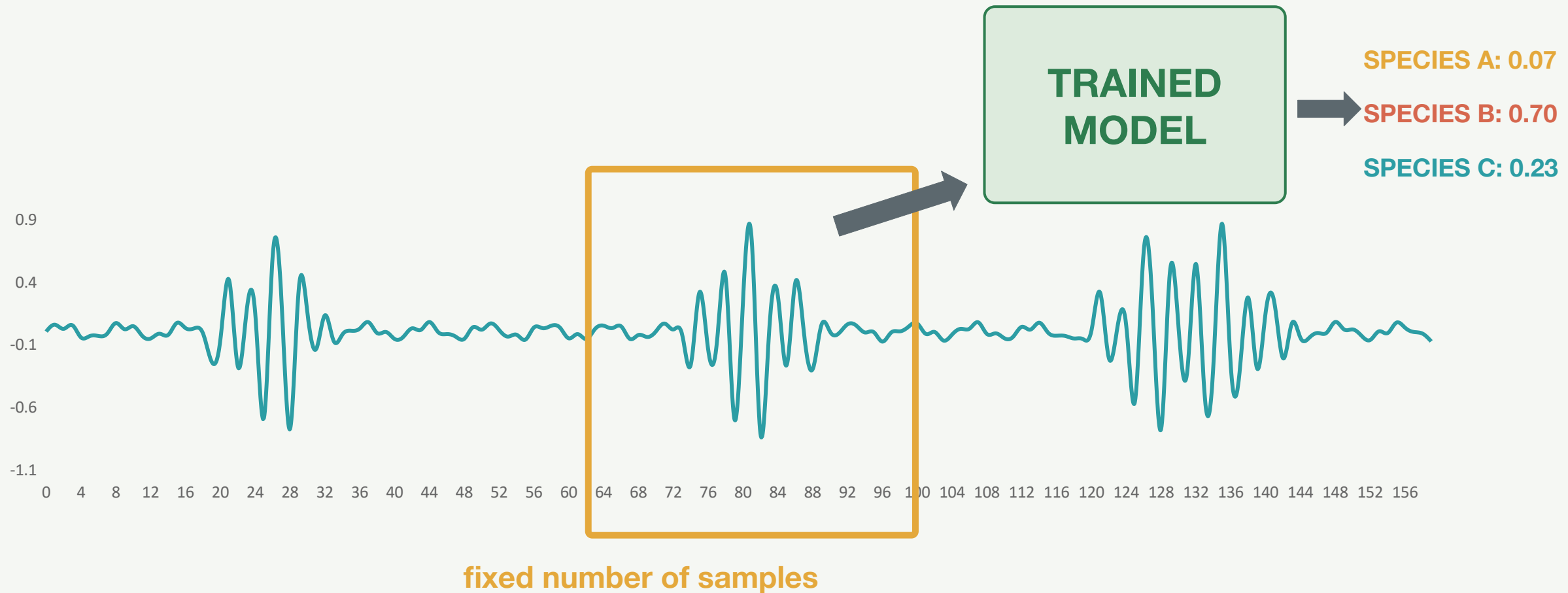
Continuous Audio Streams

microphone stream captures continuously with samples showing up at different times

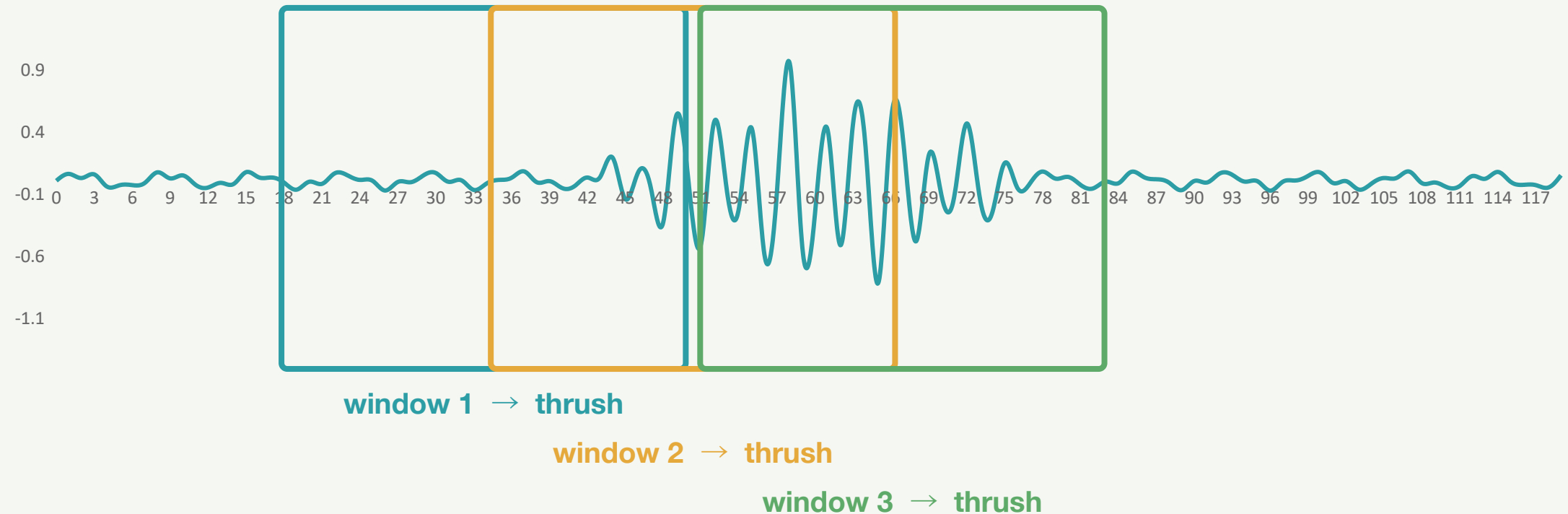


How should we feed a sample into the model?

Fixed-Size Input Windows

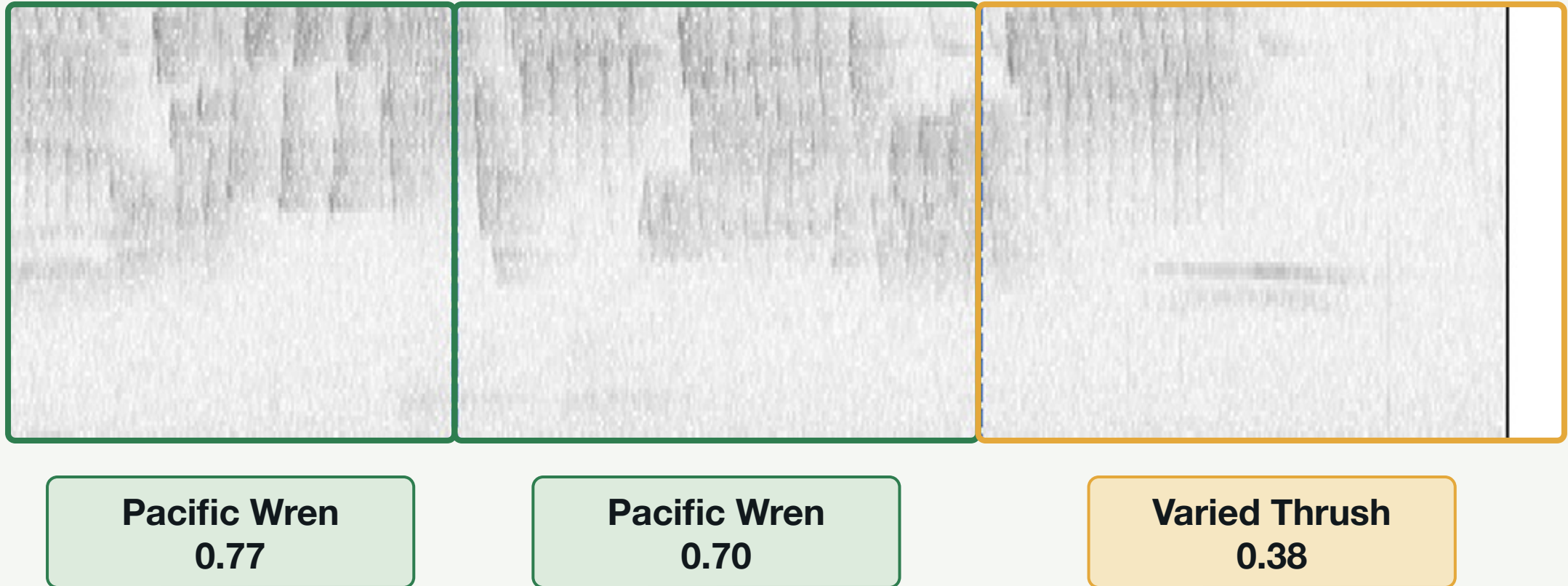


Sliding-Window Inference

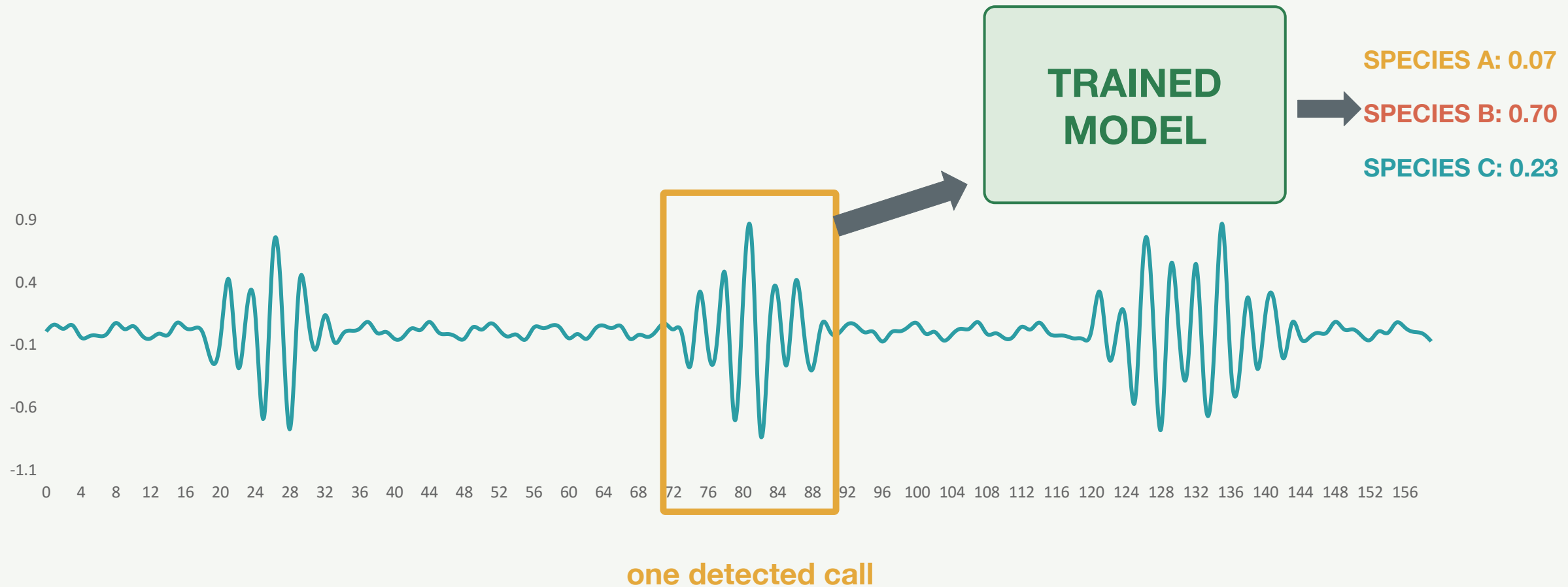


One bird call. Three predictions. How many events?

Sliding-Window Inference

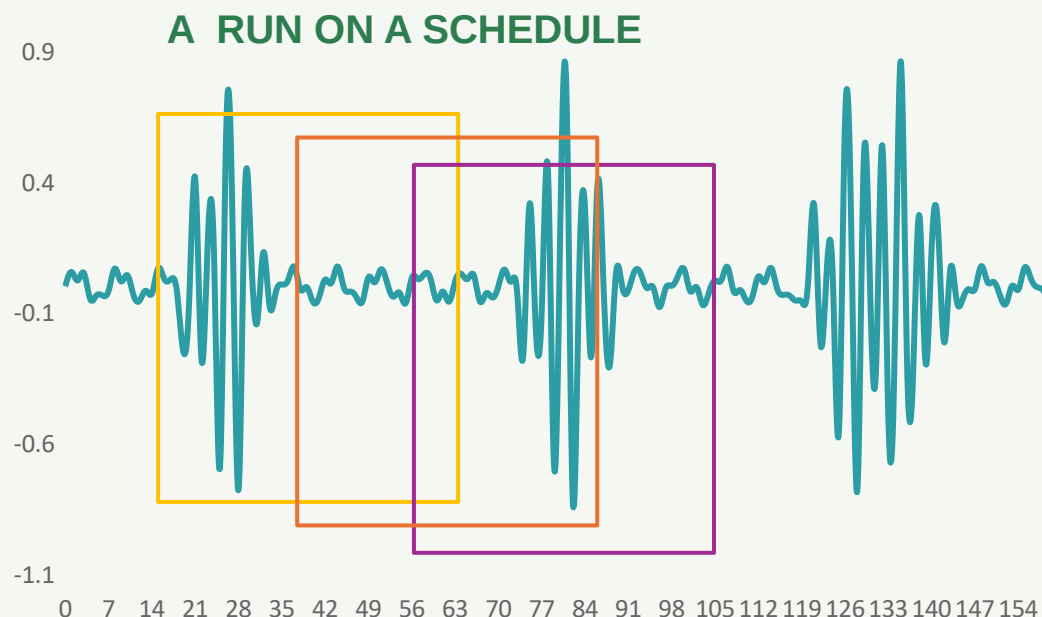


Event-Based Segmentation



A window defines what the model sees. When should it run?

The same stream supports two different inference schedules.



What must be true for each strategy to work?

Which strategy fits the system?

SLIDING WINDOWS

RUN: every fixed step

ASSUMES: only a window size

TRADEOFF: repeated + partial calls

fewer assumptions about event boundaries

DETECT → SEGMENT →
CLASSIFY

RUN: after an event detector

ASSUMES: boundaries are reliable

TRADEOFF: missed + truncated calls

fewer classifier calls

Classifier Label Set

LABEL SET

warbler

thrush

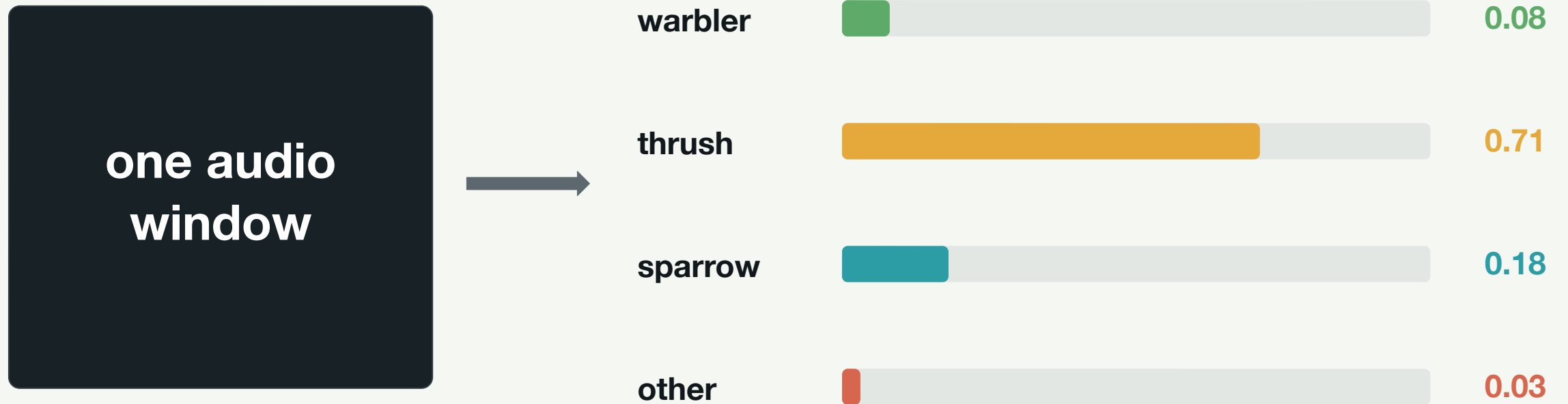
sparrow

other

A new input cannot create a fifth label.

Even “other” is a category learned from examples.

Classifier Scores



The scores are evidence under this model, not facts about the world.

Argmax Prediction

warbler

0.08

thrush

0.71

sparrow

0.18

other

0.03

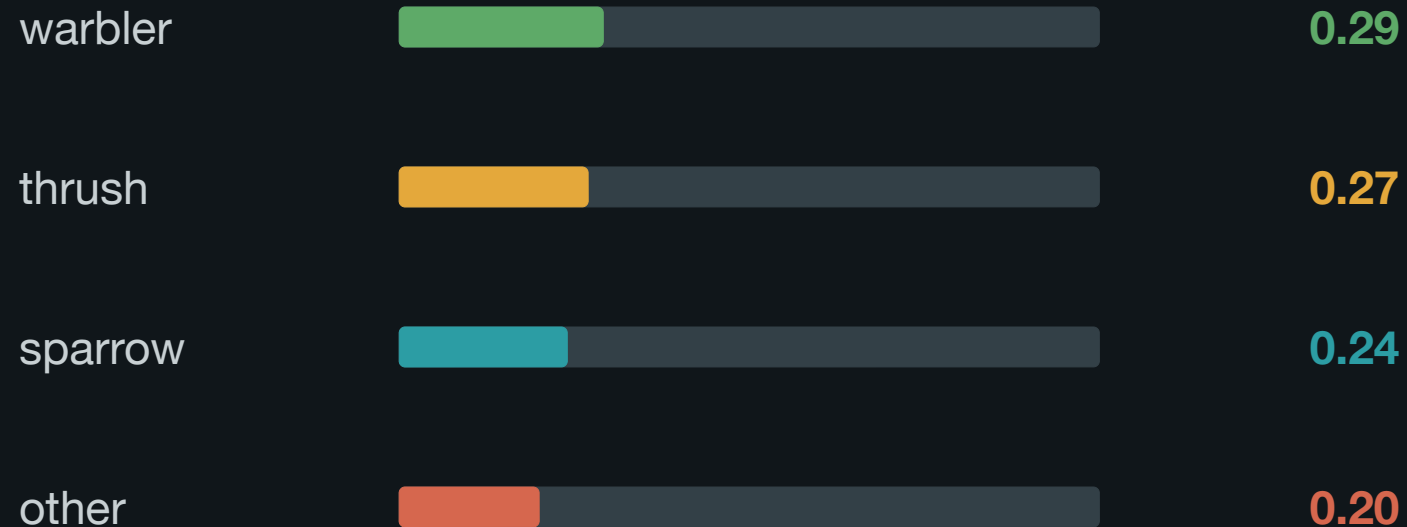
largest value



prediction: thrush

argmax = index of the largest value

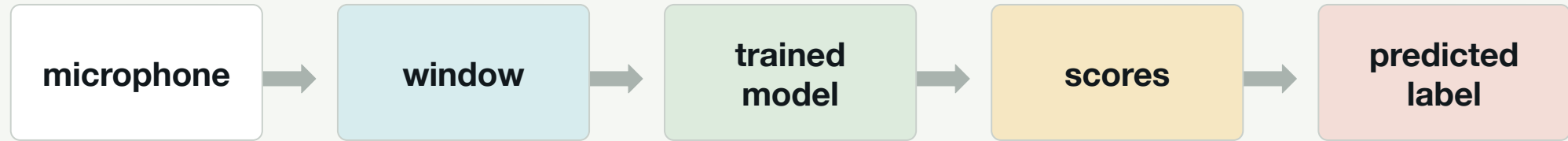
Low-Confidence Predictions



**still predicts
warbler**

Every score vector has a maximum — even for noise or an unfamiliar bird.

Model Predictions and Application Decisions



How do we know whether those predictions deserve trust?